Anuk Abeysinghe

Mechanical Engineering Undergraduate, University of Moratuwa

anuk30.github.io/Anuk-Portfolio | linkedin.com/in/anukabeysinghe | anuklithira30@gmail.com | +94 71 138 6628

Piliyandala, Sri Lanka

Profile

Third-year Mechanical Engineering undergraduate at the University of Moratuwa with hands-on experience in mechanical design, manufacturing, automation, and electric vehicle systems through rigorous academic projects and the Formula Student competition. Experienced across multidisciplinary domains including gear and machine design, fluid mechanics, CAD modeling, high-voltage accumulator systems, and electronics. Eager to apply analytical engineering design and practical problem-solving skills in an industrial environment.

Education

University of Moratuwa — **Moratuwa, Sri Lanka**
Bachelor of Science of Engineering (Hons) in Mechanical Engineering; CGPA: 3.56 / 4.00 — March 2024 – Present

Mahanama College — **Colombo 03, Sri Lanka**
- G.C.E. Advanced Level - Physical Science Stream (Z - score: 2.1567) — March 2008 – January 2022
- G.C.E. Ordinary Level: 9 As (Distinctions)

Projects

Gearbox for a Mechanical Carbon Fiber Impregnation Press — ***Ongoing***
- Designing a 3 - speed parallel axis sliding mesh spur gearbox for a mechanical carbon fibre impregnation press capable of processing 3K, 6K, and 12K tow carbon fibre.
- Determined roller operating requirements of 1.59 - 6.37 rpm and torque ratings of 30 - 75 Nm through detailed preliminary process mechanics analysis.
- Developed the complete transmission system incorporating a 10:1 chain reduction and a 1:1 spur gear drive for synchronized counter-rotating rollers.
- Selected and evaluated gears, shafts, bearings, couplings, and lubrication schemes based on calculated stress and operating requirements.

Smart Wall Painting Robot
Arduino Mega | PID Control | Mecanum Wheel Drive | Colour Sensing | Automated Spray Mechanism
- Developed an autonomous wall-painting robot utilizing 4 geared motors with a Mecanum-wheel drive controlled by an Arduino Mega for omnidirectional maneuverability.
- Implemented PID closed-loop motor control and color sensing to regulate movement and accurately distinguish painted from unpainted wall sections.
- Engineered wall-painting and letter-painting modes with a custom matrix-based spray mechanism for automated character generation.

Manufacturing & Fabrication Project
Woodworking | Welding | Metal Forming
- Manufactured a precision decorative metal and wood assembly through hands-on application of woodworking, metal forming and welding processes.
- Fabricated and assembled individual mechanical components using practical workshop measurement, marking, cutting, forming and surface finishing techniques.
- Gained practical understanding of manufacturing tolerances, workshop tooling and fabrication methodologies.

Reverse Engineering Project – Soldering Iron
- Disassembled and analyzed an electric soldering iron to investigate the construction, thermal function and assembly of key components (tip, heating element, coil, protective tube and grip enclosure).
- Conducted material identification, thermal testing, and investigated appropriate high volume manufacturing processes.
- Generated full 3D CAD parametric models and standardized 2D production engineering drawings based on physical measurements and reverse engineering analysis.

Experience

Team Falcon E – Formula Student Team, University of Moratuwa **Moratuwa, Sri Lanka**
Electronics Team Member *December 2025 – Present*

- Contributed to the design, assembly, testing, and troubleshooting of the high-voltage accumulator and electrical powertrain for a Formula Student electric race car.
- Designed and developed the vehicle's low-voltage wiring harness, including cable routing, connector selection and integration of electrical subsystems.
- Worked hands on with high-voltage battery architecture, Battery Management Systems (BMS), motor controllers and integrated electrical safety systems.
- Performed IMD testing and high-voltage electrical safety checks to verify insulation integrity and compliance with Formula Student electrical safety requirements.
- Performed PCB soldering, inspection, testing and troubleshooting for electronic control and protection circuits.
- Assisted in the diagnosis and troubleshooting of electrical system faults during vehicle testing, integrating measurements and system data to identify and resolve issues.
- Represented the **University of Moratuwa at Formula Student Spain 2026**, contributing to the team's successful completion of technical scrutineering and dynamic safety tests.

Technical Skills Summary

- **CAD & Design Software:** SolidWorks, AutoCAD, Solid Edge, Altium Designer
- **Engineering Analysis:** MATLAB, ANSYS
- **Programming Languages:** Python, C++
- **Embedded Systems & Hardware:** Arduino, ESP32, PlatformIO, 3D Printing
- **Productivity & Documentation:** MS Office Suite (Word, Excel, PowerPoint)
- **Languages:** English (Fluent), Sinhala (Native)

Memberships & Affiliations

Student Member, Institution of Engineers Sri Lanka (IESL) **Membership No: S - 32848**

References

Prof. J. R. Gamage
Senior Lecturer, Department of Mechanical Engineering
University of Moratuwa, Sri Lanka
gamagejr@uom.lk
+94 71 487 1303